

SAMPLE EFFICIENT EXPERIENCE REPLAY IN NON-STATIONARY ENVIRONMENTS

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ABSTRACT

Reinforcement learning (RL) in non-stationary environments presents significant challenges as distribution shifts in transition dynamics and rewards rapidly invalidate previously gathered experience. Traditional Experience replay (ER) methods that prioritize solely by temporal difference error (TD-error) conflate policy-induced distribution shift with exogenous environment changes, yielding suboptimal sampling and slow adaptation. To address this limitation, we introduce the Discrepancy of Environment (DoE), a metric that isolates the impact of environment transitions on value estimation. Building on DoE, we further propose the Discrepancy of Environment Prioritized Experience Replay (DEER), which jointly prioritizes samples by policy evolution and detected environmental variation. DEER employs a binary change-point detector to signal shifts and, conditional on detection, switches between pre- and post-shift prioritization tailored to the current dynamics. This design improves sample efficiency and stabilizes learning under dynamic conditions. Across four non-stationary benchmarks, DEER improves off-policy performance by 11.54% on average relative to the strongest ER baseline.

Index Terms— Reinforcement learning, non-stationary environment, experience replay, off-policy algorithm

1. INTRODUCTION

Reinforcement learning (RL) has emerged as a powerful framework for sequential decision-making and has been widely adopted in diverse practical applications [1, 2, 3, 4, 5]. However, real-world environments are inherently non-stationary, with time-varying transition dynamics and reward

signals [6, 7, 8], which degrade the performance of policies due to distribution shift. Therefore, developing adaptive policies is imperative to effectively address the complexity and unpredictability inherent in practical environments.

Off-policy RL addresses the challenge of sparse and expensive sampling in high-dimensional continuous state-action spaces by leveraging historical transitions through experience replay (ER) [9]. Recent advancements have introduced prioritized ER, which samples transitions according to their temporal difference errors (TD-errors), measuring the discrepancy between predicted and actual returns [10, 11]. By allocating higher sampling probabilities to transitions with larger TD-errors, prioritized ER accelerates convergence and improves sample efficiency, particularly in stationary environments.

However, in non-stationary environments, historical experiences quickly become outdated, which can undermine effective sample selection and mislead learning [12]. Most existing methods [13, 14] overlook the negative impact of obsolete transitions, especially as TD-error is influenced by both environmental changes and policy updates. After the value function adapts to a new environment, transitions collected before the change often show higher TD-errors, and prioritizing these can amplify irrelevant experiences, reducing training efficiency and performance. Similar issues arise with reward-based [15] or frequency-based sampling [16]. Consequently, prioritization strategies that solely focus on policy improvement are insufficient for non-stationary settings, as they do not accurately reflect the current utility of historical experiences.

To address the challenges posed by non-stationary environments, we first introduce the Discrepancy of Environment (DoE), a principled metric that quantifies the impact of environmental dynamics changes on transitions. Specifically, DoE measures the deviation between action-value functions before and after a change in environment dynamics, excluding the influence of policy improvement. This allows DoE to precisely attribute value function variations to underlying en-

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vironmental factors. Building upon DoE, we further propose the Discrepancy of Environment Prioritized Experience Replay (DEER), a sample-efficient replay framework that adaptively prioritizes experience samples for both policy optimization and environment adaptation. DEER employs a binary classifier to detect dynamic changes by estimating reward sequence distributions across adjacent time windows and applies differentiated prioritization strategies to transitions before and after detected changes. Pre-change transitions with lower DoE are prioritized as more relevant, while post-change transitions are ranked based on a hybrid of TD-error and real-time DoE-based density differences. This approach maintains replay buffer diversity and dynamically allocates sampling priorities to address the agent’s learning requirements. Extensive experiments on four non-stationary Mujoco benchmarks demonstrate that, compared to state-of-the-art ER baselines, DEER improves the performance of off-policy algorithms by 11.54%. Furthermore, under extreme non-stationarity, DEER achieves an 22.53% performance improvement for off-policy algorithms over the best baseline.

2. RELATED WORK

ER enhances sample efficiency and learning stability by storing and resampling past experiences [17]. The most common proportional sampling method involves prioritizing transitions based on TD-error, which has demonstrated performance improvements in many RL tasks [16, 10]. Other studies utilize alternative metrics as sampling priorities. PSER assigns high priority to significant transitions and increases the priority of previous transitions that resulted in significant transitions [14]. ReF-ER classifies transitions as “near-policy” or “far-policy” based on the ratio of the current policy to the past policy, and only samples near-policy transitions [18]. AER is a sampling method that prioritizes transitions in the replay buffer based on their similarity to the current state [16]. HER [19] and its variants [20, 21] improve sample efficiency in sparse reward tasks by relabeling transitions to generate sufficient reward feedback. RB-PER assigns high priority to transitions that have been sampled less frequently, ensuring diversity in sampling and thereby adapting to non-stationary environments [22].

3. METHODOLOGY

3.1. Problem Formalization

RL in non-stationary environments is formulated as a family of Markov decision processes (MDPs) [23], denoted by $\{\langle \mathcal{S}, \mathcal{A}, P_i, R_i, T_i, \gamma \rangle\}_{i=0}^{\infty}$. Here, $\mathcal{S}$ is the state space, and $\mathcal{A}$ is the action space. At each time step t , the agent selects an action a_t according to the policy $\pi(a_t | s_t)$, transitions to the next state s_{t+1} according to the state-transition probability $P(s_{t+1} | s_t, a_t)$, and receives a reward $r_t = R(s_t, a_t)$. The state-transition probability in non-stationary environments is

defined as:

$$P(s_{t+1} | s_t, a_t) = \begin{cases} P_0(s_{t+1} | s_t, a_t), & 0 \leq t \leq T_0 \\ \dots \\ P_i(s_{t+1} | s_t, a_t), & T_{i-1} < t \leq T_i \\ \dots \end{cases} \quad (1)$$

where T_i denotes the time step at which the environment dynamics change. The reward function of non-stationary environments is defined as:

$$R(s_t, a_t) = \begin{cases} R_0(s_t, a_t), & 0 \leq t \leq T_0 \\ \dots \\ R_i(s_t, a_t), & T_{i-1} < t \leq T_i \\ \dots \end{cases} \quad (2)$$

The objective of the agent is to learn a policy π to maximize the expected discounted return $\mathbb{E}_{\pi} [\sum_{k=0}^{\infty} \gamma^k r_{t+k}]$, where γ denotes the discount factor.

3.2. Discrepancy of Environment (DoE)

To prioritize samples that enable rapid adaptation in non-stationary MDPs, we assign higher replay priority to transitions indicative of dynamics shifts. We consider a trajectories $\tau = [s_0, a_0, s_1, a_1, \dots]$ generated by a policy π that has converged under the state-transition probability P_0 and reward function R_0 . Since the state-transition probability captures the environment’s dynamics, changes in these trajectories provide reliable signals of dynamic shifts:

$$P(\tau) = P(s_0) \prod_{t=0}^{\infty} \pi(a_t | s_t) P(s_{t+1} | s_t, a_t), \quad (3)$$

where $P(s_0)$ denotes the initial state distribution. For a converged policy and fixed initial state distribution, variations in the environment’s transition dynamics directly alter the trajectory distribution. Since the reward sequence $\mathbf{r} = [r_0, r_1, \dots]$ is determined by the trajectory and reward function, shifts in environment dynamics can be indirectly detected by monitoring changes in the distribution of observed reward sequences.

To effectively detect such changes, we propose an environmental dynamics monitoring approach based on density-ratio estimation [24], leveraging a multilayer perceptron (MLP) based binary classifier trained on reward sequences. Specifically, at each time step t , we construct two adjacent sliding windows: a reference window $\tilde{\mathbf{r}}_{rf} = \{\mathbf{r}_{i:j} | t - 2m + 1 \leq i < j \leq t - m\}$, labeled as $l = 0$, and a test window $\tilde{\mathbf{r}}_{te} = \{\mathbf{r}_{i:j} | t - m < i < j \leq t\}$, labeled as $l = 1$, where m denotes the window size. The binary classifier is trained to distinguish samples from the reference and test windows, thereby implicitly estimating the density ratio between their distributions. Formally, the classifier models the conditional distribution as follows:

$$P(\mathbf{r} | f(\mathbf{r}) = l) = \begin{cases} P_{rf}(\mathbf{r}) & \text{if } l = 0 \\ P_{te}(\mathbf{r}) & \text{if } l = 1 \end{cases}, \quad (4)$$

where P_{rf} and P_{te} denote the density functions of the reference and test windows, respectively, and f represents the binary classifier. The MLP parameters are optimized by minimizing the following cross-entropy loss:

$$\mathcal{L}(f) = -\frac{1}{|\tilde{\mathbf{r}}_{rf}|} \sum_{\mathbf{r} \in \tilde{\mathbf{r}}_{rf}} \log(1 - f(\mathbf{r})) - \frac{1}{|\tilde{\mathbf{r}}_{te}|} \sum_{\mathbf{r} \in \tilde{\mathbf{r}}_{te}} \log f(\mathbf{r}). \quad (5)$$

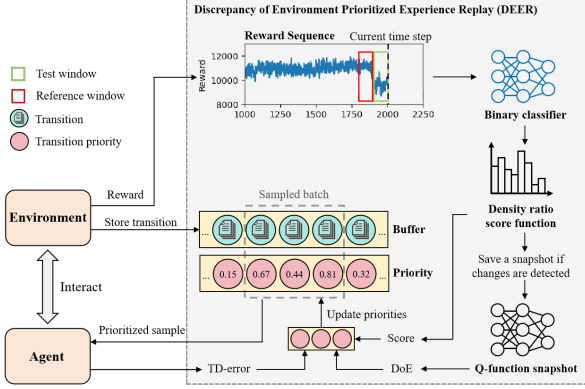


Fig. 1: The workflow of DEER.

A change point at time step $t - m$ is identified if the density ratio score $S(\tilde{\mathbf{r}}_{te}, \tilde{\mathbf{r}}_{rf})$ exceeds a predefined threshold μ (i.e., $S \geq \mu$). Specifically, we adopt the Jensen-Shannon divergence as the density ratio score function, formulated as:

$$S(\tilde{\mathbf{r}}_{te}, \tilde{\mathbf{r}}_{rf}) = \log 2 + \frac{1}{2n_{te}} \sum_{\mathbf{r} \in \tilde{\mathbf{r}}_{te}} \log P(\mathbf{r} | f(\mathbf{r}) = 1) + \frac{1}{2n_{rf}} \sum_{\mathbf{r} \in \tilde{\mathbf{r}}_{rf}} \log P(\mathbf{r} | f(\mathbf{r}) = 0). \quad (6)$$

To quantify the impact of environmental dynamics changes on state-action value estimation, we introduce the Discrepancy of Environment (DoE) metric. For a given state-action pair (s_k, a_k) , DoE measures the difference in Q-function values before and after an environmental change, defined as:

$$\begin{aligned} DoE(s_k, a_k) &= \mathbb{E}_{\tau \sim (P_i, \pi)} \left[\sum_{t=k}^{\infty} \gamma^{t-k} R_i(s_t, a_t) \right] \\ &\quad - \mathbb{E}_{\tau \sim (P_{i-1}, \pi)} \left[\sum_{t=k}^{\infty} \gamma^{t-k} R_{i-1}(s_t, a_t) \right] \\ &= Q_i(s_k, a_k) - Q_{i-1}(s_k, a_k), \end{aligned} \quad (7)$$

where Q_{i-1} denotes the Q-function from the previous state-transition probability P_{i-1} and reward function R_{i-1} , while Q_i represents the current Q-function under the new state-transition probability P_i and reward function R_i .

3.3. Discrepancy of Environment Prioritized Experience Replay (DEER)

We further propose DEER, as illustrated in Figure 1. For pre-change transitions, DEER prioritizes samples with a low DoE, as these are less perturbed and remain informative after the change. The priority assigned to a transition collected at time step k ($k \leq T_{i-1}$) is defined as:

$$p_k = 2\sigma(-|DoE(s_k, a_k)|), \quad (8)$$

where p_k denotes the priority of the transition collected at time step k and σ denotes the sigmoid normalization. For post-change transitions ($k > T_{i-1}$), DEER employs a hybrid metric sampling strategy that integrates both the TD-error and DoE, dynamically weighted by the real-time density ratio score. The density ratio score (Eq. 6)

Offset	0%	50%	200%
DEER	11894.85 $\pm$ 182.36	11789.66 $\pm$ 283.81	9856.27 $\pm$ 477.04
PER	11834.68 $\pm$ 244.31	10266.77 $\pm$ 327.79	5204.22 $\pm$ 763.55
RB-PER	11754.29 $\pm$ 202.61	10506.37 $\pm$ 364.37	7332.36 $\pm$ 318.96
CER	11975.13 $\pm$ 256.47	9428.88 $\pm$ 199.17	8043.76 $\pm$ 467.35
LA3P	11895.44 $\pm$ 267.72	11024.15 $\pm$ 787.15	6614.79 $\pm$ 285.27

Table 1: Average rewards over 10^5 steps post-change for SAC+HalfCheetah under different levels of non-stationarity.

quantifies the shift in the reward distribution, where a high score indicates that the agent needs to adapt to the altered environment. In such scenarios, transitions with elevated DoE are prioritized to expedite adaptation. Conversely, when the density ratio score is low, sampling emphasizes transitions with higher TD-error to facilitate policy refinement. This adaptive prioritized sampling mechanism is formally defined as follows:

$$p_k = (1 - S(\tilde{\mathbf{r}}_{te}, \tilde{\mathbf{r}}_{rf})) (2\sigma(|TD(s_k, a_k, r_k, s_{k+1})|) - 1) + S(\tilde{\mathbf{r}}_{te}, \tilde{\mathbf{r}}_{rf}) (2\sigma(|DoE(s_k, a_k)|) - 1), \quad (9)$$

where $TD(s_k, a_k, r_k, s_{k+1}) = r_k + \gamma Q_i(s_{k+1}, a_{k+1}) - Q_i(s_k, a_k)$ denotes the TD-error. To prevent overfitting due to repeated sampling of high-priority transitions, all sampling probabilities are normalized as:

$$P(k) = \frac{p_k^\alpha}{\sum_i p_i^\alpha}, \quad (10)$$

where $\alpha \in (0, 1]$ controls the degree of prioritization. To correct the sampling-induced bias in Q-function estimation, importance sampling weights are introduced:

$$w_k = \frac{1}{(N \cdot P(k))^\beta \cdot \max_i w_i}, \quad (11)$$

where w_k denotes the update weight of the transition k , and N denotes replay buffer size and $\beta \in (0, 1]$ controls the extent of bias correction.

4. EVALUATION

4.1. Experimental Setup

4.1.1. Environment

We evaluate our proposed integration with the SAC algorithm [25] on four standard MuJoCo Gymnasium continuous control tasks: Ant-v4, HalfCheetah-v4, Hopper-v4, and Inverted Double Pendulum-v4 (ID-Pendulum-v4). To simulate non-stationary environments, we introduce offsets to friction and joint damping coefficients—set to the largest values that still permit SAC with uniform sampling to converge.

4.1.2. Baselines

We compare several experience replay (ER) methods, including PER [13], RB-PER [22], CER [26], LA3P [27], and our proposed DEER. Training is run for 5×10^5 steps on ID-Pendulum and 1×10^6 steps on the other tasks, with non-stationarity introduced halfway.

4.1.3. Hyperparameters

Comparative results are averaged over 5 independent runs. All networks use two hidden layers of 256 units, a learning rate of 1×10^{-3} , discount factor of 0.99 and replay buffer size of 1×10^6 . For DEER, we set $\alpha = 0.6$, $\beta = 0.4$, and use a binary classifier (MLP with two

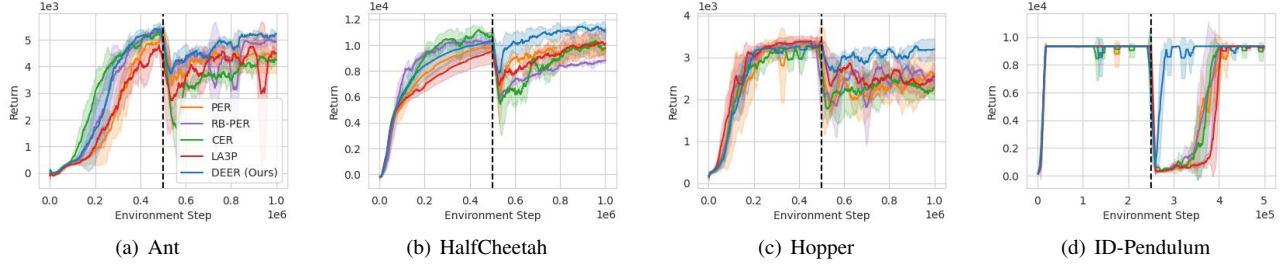


Fig. 2: Sample efficiency comparison of SAC across four tasks. The dashed line marks when the environment dynamics change.

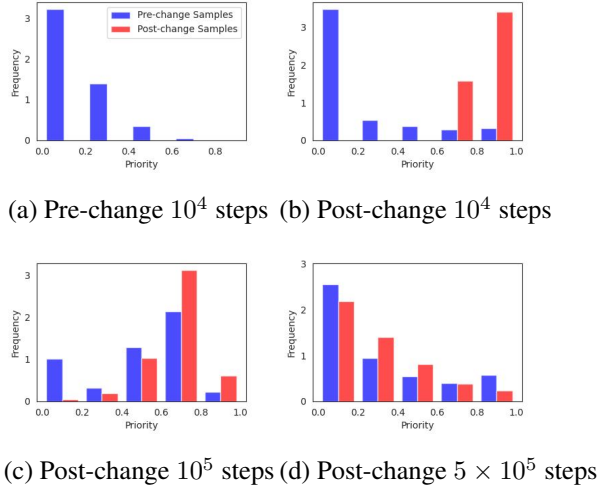


Fig. 3: Priority distribution of pre- and post-change samples.

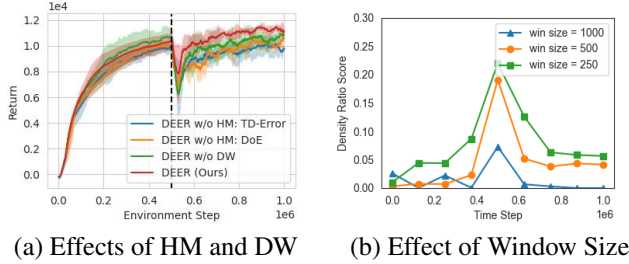


Fig. 4: Ablation Results.

128 units hidden layers) trained for up to 50 iterations. The detection window is 500, with 10 samples per window (each containing 50 data points), and the detection threshold is set to 0.2.

4.2. Experimental Results and Analysis

4.2.1. Comparative Results

Figure 2(a)–2(d) present episode-return curves for SAC combined with several state-of-the-art ER, including PER [13], RB-PER [22], CER [26], LA3P [27], and our proposed DEER. Across settings, DEER achieves higher cumulative returns, with smaller post-change degradation and faster recovery than baselines, evidencing stronger adaptability to dynamic shifts.

4.2.2. Sampling Behavior Analysis

Figure 3 shows the priority distribution of samples at different training stages. Post-change transitions have higher priority than pre-change ones, especially early after environmental shifts. As training progresses and density ratio scores decrease, the distribution gradually returns to its pre-change state. This reflects DEER’s balance between exploiting recent experience for rapid adaptation and reusing historical experience for stability.

4.2.3. Ablation Study

Figure 4(a) shows the ablation study in HalfCheetah with SAC on two key designs: hybrid metric sampling (HM) and dynamic weighting (DW). In DEER w/o DW, the density-ratio score is used for change detection, but not for weighting priorities. In DEER w/o HM, we evaluate two single-metric variants using only TD error or only DoE for priority. HW and DW markedly improve post-change performance; removing DW or relying on a single metric reduces adaptability and sample efficiency. These results indicate that combining DoE and TD error is critical for robust priority estimation, while DW further enhances adaptation to evolving dynamics.

4.2.4. Effect of Window Size

Figure 4(b) shows an analysis of varying time-window sizes. Window sizes of 250, 500, and 1000 effectively detect changes. Smaller windows react more quickly, yielding larger score jumps and lower detection latency, but can be sensitive to minor fluctuations. In practice, a window size of 500 works well in all tested tasks.

4.2.5. Robustness

Table 1 reports average rewards over 100 episodes after a change in SAC+HalfCheetah under extreme (200% offset), mild (50% offset), and stationary (0% offset) settings; the standard setting (100% offset) corresponds to Figure 2(b). DEER substantially outperforms baselines under extreme non-stationarity, achieving 22.53% higher rewards than the best competitor at 200% offset, highlighting its strong adaptability to drastic shifts. Under mild non-stationarity, DEER retains a modest advantage. Under stationarity, all methods perform comparably, indicating DEER does not degrade performance in stationarity environments.

5. CONCLUSION

We propose DEER, a hybrid metric prioritization framework that adaptively adjusts sampling weights in response to non-stationary environmental dynamics. DEER prioritizes informative samples in accordance with real-time environmental changes using multiple relevance metrics, thereby enhancing sample efficiency.

6. REFERENCES

- [1] Zheng Lin, Wei Wei, Zhe Chen, Chan-Tong Lam, Xianhao Chen, Yue Gao, and Jun Luo, “Hierarchical Split Federated Learning: Convergence Analysis and System Optimization,” *IEEE Trans. Mobile Comput.*, 2025.
- [2] Chen Tang, Ben Abbatematteo, Jiaheng Hu, Rohan Chandra, Roberto Martín-Martín, and Peter Stone, “Deep reinforcement learning for robotics: A survey of real-world successes,” in *Proc. AAAI*, 2025, vol. 39, pp. 28694–28698.
- [3] Zongyuan Zhang, Tianyang Duan, Zheng Lin, Dong Huang, Zihan Fang, Zekai Sun, Ling Xiong, Hongbin Liang, Heming Cui, Yong Cui, and Yue Gao, “Robust Deep Reinforcement Learning in Robotics via Adaptive Gradient-Masked Adversarial Attacks,” in *Proc. IEEE/RSJ IROS*, 2025, pp. 20418–20425.
- [4] Zheng Lin, Guanqiao Qu, Qiyuan Chen, Xianhao Chen, Zhe Chen, and Kaibin Huang, “Pushing Large Language Models to the 6G Edge: Vision, Challenges, and Opportunities,” *IEEE Communication Magazine*, 2025.
- [5] Zihan Fang, Zheng Lin, Zhe Chen, Xianhao Chen, Yue Gao, and Yuguang Fang, “Automated Federated Pipeline for Parameter-Efficient Fine-Tuning of Large Language Models,” *IEEE Trans. Mobile Comput.*, 2025.
- [6] Zheng Lin, Yuxin Zhang, Zhe Chen, Zihan Fang, Cong Wu, Xianhao Chen, Yue Gao, and Jun Luo, “LEO-Split: A Semi-Supervised Split Learning Framework over LEO Satellite Networks,” *IEEE Trans. Mobile Comput.*, 2025.
- [7] Khimya Khetarpal, Matthew Riemer, Irina Rish, and Doina Precup, “Towards continual reinforcement learning: A review and perspectives,” *J. Artif. Intell. Res.*, vol. 75, pp. 1401–1476, 2022.
- [8] Zheng Lin, Guanqiao Qu, Wei Wei, Xianhao Chen, and Kin K Leung, “Adaptsfl: Adaptive Split Federated Learning in Resource-Constrained Edge Networks,” *IEEE Trans. Netw.*, 2025.
- [9] Denis Yarats, Amy Zhang, Ilya Kostrikov, Brandon Amos, Joelle Pineau, and Rob Fergus, “Improving sample efficiency in model-free reinforcement learning from images,” in *Proc. AAAI*, 2021, vol. 35, pp. 10674–10681.
- [10] Hu Li, Xuezhong Qian, and Wei Song, “Prioritized experience replay based on dynamics priority,” *Sci. Rep.*, vol. 14, no. 1, pp. 6014, 2024.
- [11] Zongyuan Zhang, Tianyang Duan, Zheng Lin, Dong Huang, Zihan Fang, Zekai Sun, Ling Xiong, Hongbin Liang, Heming Cui, and Yong Cui, “State-Aware Perturbation Optimization for Robust Deep Reinforcement Learning,” *IEEE Trans. Mobile Comput.*, vol. 25, no. 1, pp. 992–1008, 2026.
- [12] Ali Rahimi-Kalahroudi, Janarthanan Rajendran, Ida Momennejad, Harm van Seijen, and Sarath Chandar, “Replay Buffer with Local Forgetting for Adapting to Local Environment Changes in Deep Model-Based Reinforcement Learning,” in *Proc. CoLLA*, 2023, pp. 21–42.
- [13] Tom Schaul, John Quan, Ioannis Antonoglou, and David Silver, “Prioritized Experience Replay,” *arXiv preprint arXiv:1511.05952*, 2015.
- [14] Marc Brittain, Josh Bertram, Xuxi Yang, and Peng Wei, “Prioritized sequence experience replay,” *arXiv preprint arXiv:1905.12726*, 2019.
- [15] Xi Cao, Huaiyu Wan, Youfang Lin, and Sheng Han, “High-value prioritized experience replay for off-policy reinforcement learning,” in *Proc. IEEE ICTAI*, 2019, pp. 1510–1514.
- [16] Peiquan Sun, Wengang Zhou, and Houqiang Li, “Attentive experience replay,” in *Proc. AAAI*, 2020, vol. 34, pp. 5900–5907.
- [17] Long-Ji Lin, “Self-improving reactive agents based on reinforcement learning, planning and teaching,” *Mach. Learn.*, vol. 8, pp. 293–321, 1992.
- [18] Guido Novati and Petros Koumoutsakos, “Remember and forget for experience replay,” in *Proc. ICML*, 2019, pp. 4851–4860.
- [19] Marcin Andrychowicz, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, OpenAI Pieter Abbeel, and Wojciech Zaremba, “Hindsight experience replay,” *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017.
- [20] Yongle Luo, Yuxin Wang, Kun Dong, Qiang Zhang, Erkan Cheng, Zhiyong Sun, and Bo Song, “Relay Hindsight Experience Replay: Self-guided continual reinforcement learning for sequential object manipulation tasks with sparse rewards,” *Neurocomputing*, vol. 557, pp. 126620, 2023.
- [21] Erdi Sayar, Vladislav Vintaykin, Giovanni Iacca, and Alois Knoll, “Hindsight Experience Replay with Evolutionary Decision Trees for Curriculum Goal Generation,” in *Proc. EvoApplications*, 2024, pp. 3–18.
- [22] Derek Li, Andrew Jacobsen, and Adam White, “Revisiting experience replay in non-stationary environments,” in *Proc. AAMAS*, 2021.
- [23] Tianyang Duan, Zongyuan Zhang, Zheng Lin, Yue Gao, Ling Xiong, Yong Cui, Hongbin Liang, Xianhao Chen, Heming Cui, and Dong Huang, “Rethinking adversarial attacks in reinforcement learning from policy distribution perspective,” in *Proc. IEEE ICASSP*, 2025, pp. 1–5.
- [24] Jonathan Hillman and Toby Dylan Hocking, “Optimizing roc curves with a sort-based surrogate loss function for binary classification and changepoint detection,” *arXiv preprint arXiv:2107.01285*, 2021.
- [25] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine, “Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor,” in *Proc. ICML*, 2018, pp. 1861–1870.
- [26] Shangdong Zhang and Richard S Sutton, “A deeper look at experience replay,” *arXiv preprint arXiv:1712.01275*, 2017.
- [27] Baturay Saglam, Furkan B Mutlu, Dogan C Cicek, and Suleyman S Kozat, “Actor prioritized experience replay,” *J. Artif. Intell. Res.*, vol. 78, pp. 639–672, 2023.